Ella Standerford

Honors Student in Computer Science & Mathematics

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SUMMARY

Honors student pursuing degrees in Computer Science and Mathematics with experience in software development, data analysis, and scientific computing. Maintains a 4.0 GPA while contributing to atmospheric science research at NASA Langley Research Center and developing independent software projects in Java and Python.

EDUCATION

Northern Michigan University | Expected Graduation: May 2029
B.S. in Computer Science & Mathematics,
Minor in Business Administration
Honors Program Student | GPA: 4.0

Negaunee High School
Co-Salutatorian | GPA: 4.32

TECHNICAL SKILLS

Language & Tools: Java, Python, Git/GitHub, HTML/CSS
Data & Analytics: Matplotlib, Pandas, Data Visualization, Statistical Analysis
Professional: Microsoft Office, Project Management, Technical Documentation

PROFESSIONAL EXPERIENCE

Analytical Mechanics Associates (AMA) | Hampton, VA
***Surface Radiation, Aerosol & Meteorological Measurements Intern* | June 2026-August 2026**

- Analyze complex atmospheric radiation and meteorological datasets using Python to support ongoing NASA Langley Research Center climate research.
- Design and develop interactive graphical representations of data using Python (Matplotlib/Pandas) for upcoming deployment to the official project webpage.

NMU Network Infrastructure Center | Marquette, MI
Student Employee* | *2025-Present

- Install and program campus-wide phone systems and provide front-line technical support to faculty and staff
- Answer, route, and document service calls to ensure efficient, clear IT communication

Irontown Pasties | Negaunee, MI
Employee* | *2022-2025

- Managed fast-paced food preparation, customer transactions, and nightly cash reconciliation
- Assisted in training new staff and supported daily operations to maintain high service standards

LEADERSHIP & AWARDS

Business Professionals of America (BPA): President (2024-2025), VP (2023-2024). Managed chapter operations; achieved Multiple Top 10 National Conference Placements in Microsoft Events

Key Club and NHS: Key Club President (2024-2025) and VP; National Honor Society Treasurer (2024-2025). Organized community service and managed chapter funds.

NMU Campus Engagement: Participant in Student Leaders Fellowship Program (SLFP), Superior Edge, and Honors Program

Award: National Center for Women & Information Technology (NCWIT) Regional Affiliate Award (2025)

TECHNICAL PROJECTS

Atmospheric Data Visualization Dashboard (Python, Pandas, Matplotlib)

- Built a dashboard to clean, process, and visualize massive environmental and atmospheric datasets
- Created interactive graphics to identify critical trends in weather and radiation metrics

Isle Royale Trail Planner (Python, HTML, CSS)

- Developed a full-stack Flask app utilizing Dijkstra's Algorithm to dynamically calculate and map backcountry trail routes
- Integrated responsive UI features, including an editable meal plan tracker and packing list

Interactive Java Maze Game (Java, JavaFX)

- Built a desktop maze game applying object-oriented Java principles and the JavaFX rendering library
- Engineered custom collision logic and event-driven keyboard listeners for real-time player navigation

Personal Portfolio Website (HTML, CSS, Git)

- Designed and deployed a responsive personal website via GitHub Pages to host professional project documentation